Problem 1: 1. Batch

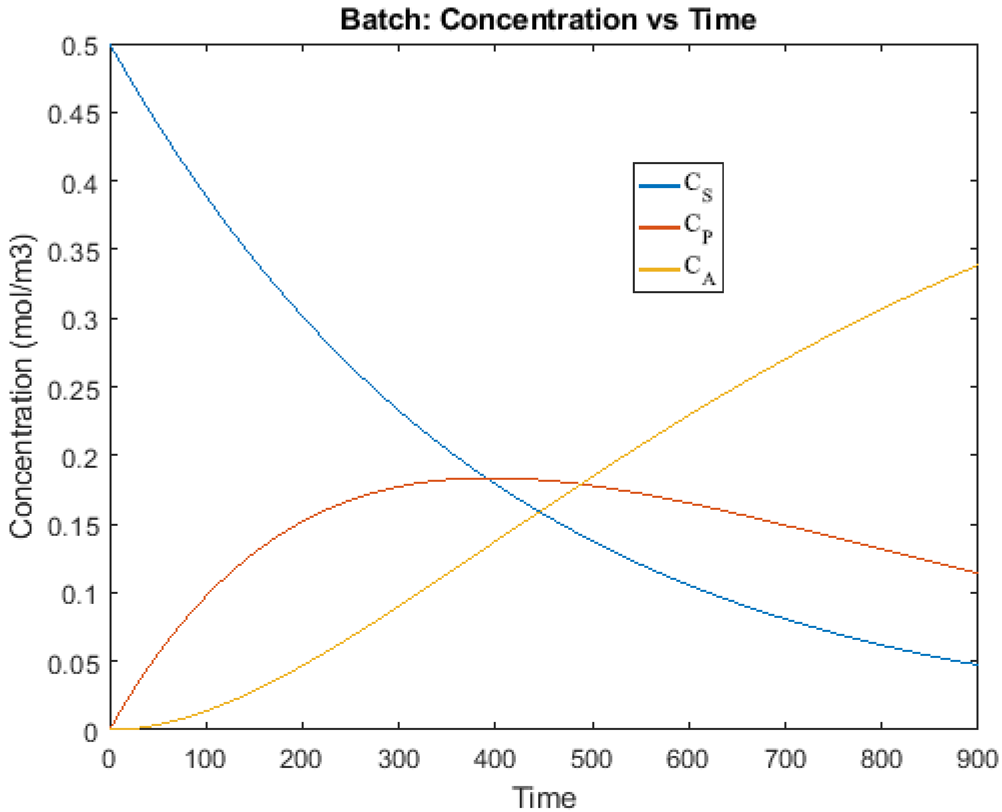
```
function hw21

t=0:3:900;
[t,C]= ode45( @ode2,t, [0.5 0 0]);
figure(1)
plot (t,C);
title(' Batch: Concentration vs Time');
xlabel('Time'); ylabel('Concentration (mol/m3)');
%C is a 9000x3 matrix
figure(2)
Selectivity = C(:,2)./(0.5-C(:,1));
plot(t,Selectivity);
title('Selectivity');
xlabel('Time'); ylabel('Selectivity');


end

function dc=ode2(t,C)
alpha=[-1 1 0
        0 -1 1];
k1=0.015;
km=5.53;
k2=0.0026;
S=C(1);
P=C(2);
A=C(3);
r1=k1*S/(km+S);
r2=k2*P;
r=[r1;r2];
dc=alpha'*r;
%alpha is 2x3 matrix. alpha' is 3x2 matrix
%r is 2x1 matrix.
%dc is (3x2)x(2x1) = 3x1 matrix.
end
```

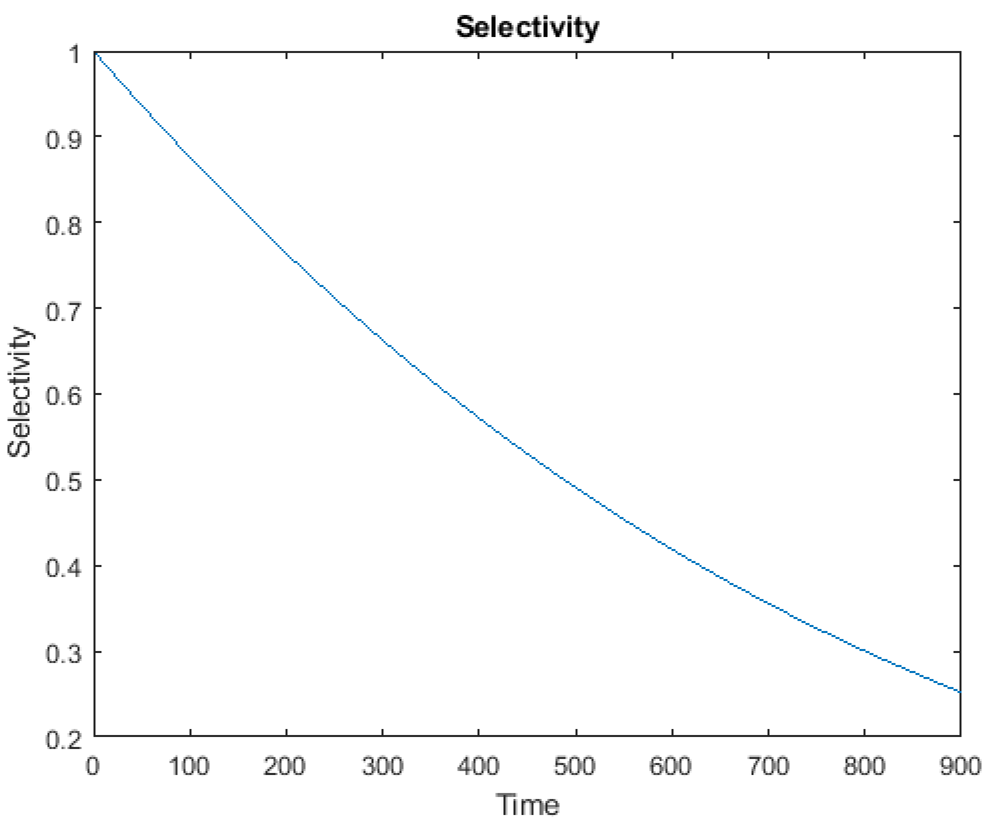
Batch reactor model equation

$$\frac{d\underline{C}}{dt} = \underline{\underline{\alpha}}^T \underline{r}$$

$$\frac{d\underline{\xi}}{dt} = \underline{r}$$

$$C_j = C_{jo} + \sum_i \alpha_{ij}\ \xi_i$$


Batch: Concentration vs Time
Concentration (mol/m3)
Time
C_S
C_P
C_A
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
100
200
300
400
500
600
700
800
900


Selectivity
Selectivity
Time
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
100
200
300
400
500
600
700
800
900

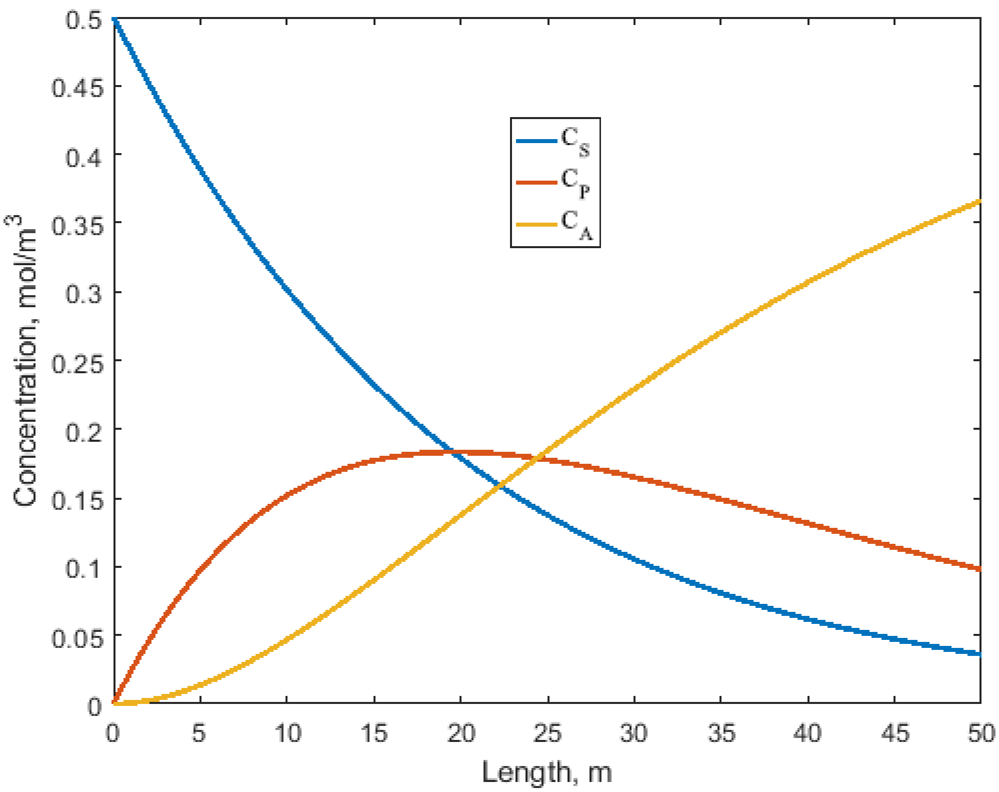
Problem 1: 2. PFR

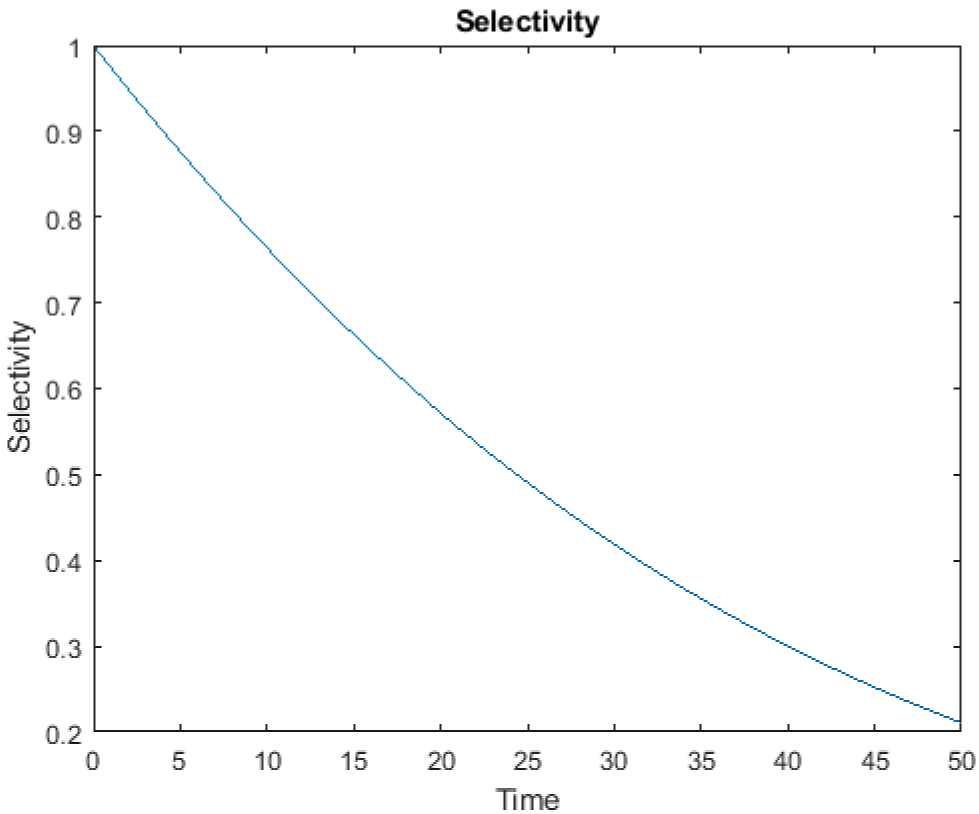
```
% PFR
clear all
Co = [0.5 0 0];
zspan = 0:0.1:50; % why Z span is to 50? shouldn't be to 5 since the units
are per cubic meter

[zsol,Csol] = ode45(@(z,C) funhw3(z,C),zspan,Co);

figure(3)
plot(zsol,Csol(:,1),'linewidth',2)
hold on
plot(zsol,Csol(:,2),'linewidth',2)
plot(zsol,Csol(:,3),'linewidth',2)
xlabel('Length, m')
ylabel('Concentration, mol/m^3')

figure(4)
Selectivity = Csol(:,2)./(0.5-Csol(:,1));
plot(zsol,Selectivity);
title('Selectivity');
xlabel('Time'); ylabel('Selectivity');

function dC = funhw3(z,C)
dC = [];
S = C(1);
P = C(2);

k1 = 0.015;
k2 =  0.0026;
K_M =  5.53;
v = 0.05; %velocity of reactants.

r1 = k1*S/(K_M + S);
r2 = k2*P;

%dC is a column vector.
dC(1,1) = (-r1)/v;
dC(2,1) = (r1-r2)/v;
dC(3,1) = (r2)/v;

end
```

Assumptions

1) Steady state
2) 1D Cartesian co-ordinates
3) Diffusion is negligible
4) Constant velocity

$$v\frac{d\underline{\underline{C}}}{dz} = \underline{\underline{\alpha}}^T \underline{r}$$





To improve the conversion of the reactant,

1) The length of the reactor must be increased.
2) The velocity of the reactants should be decreased.

PFR is preferred as the same conversion can be obtained at smaller volume.

Batch operation – 260 s operation in 1 m^3 volume of the reactor.

PFR operation – continuous operation in a reactor of length 8m and diameter 10 cm.

Problem 2

Part A:

(i) *Formulate the differential equation for the concentration of gas A neglecting radial diffusion but of course including axial diffusion with coefficient D along the pore. [5 points]*

$$\frac{DC_A}{Dt} + \nabla J_A = -C_A \frac{D \ln \tilde{V}}{Dt}$$

$$\frac{\partial C_A}{\partial t} + V.\left[\frac{1}{r}\frac{\partial (rC_A)}{\partial r} + \frac{\partial C_A}{\partial z}\right] - \frac{1}{r}\frac{\partial}{\partial r}\left(rD\frac{\partial C_A}{\partial r}\right) = 0$$

(ii) *Identify the boundary conditions for the diffusion of the gas within the deforming pore. Neglect external mass transfer resistance. State your assumptions. [5 points]*

$$z = 0, C_A = C_{Af}$$

$$r = 0, \frac{\partial C_A}{\partial r} = 0$$

$$r = R, K(C_A - C_{A,}) = \acute{K}AC_{As}$$

(iii) *Obtain the differential equation for $r(z,t)$ representing the shape of the pore assuming C_s to be the molar density of the solid. Provide initial and boundary conditions for (z,t). [5 points]*

$$f(r,z,t) = 2\pi rz$$

$$\frac{\partial f}{\partial t} = \acute{\alpha}\hat{r}[\nabla.f]$$

$$2\pi z\frac{\partial r}{\partial t} = \acute{\alpha}\hat{r}\left[\frac{1}{r}\frac{\partial (rf)}{\partial r}\right]$$

$$2\pi z\frac{\partial r}{\partial t} = \acute{\alpha}\hat{r}\left[\frac{1}{r}\frac{\partial (rf)}{\partial r}\right]$$

$$2\pi z\frac{\partial r}{\partial t} = \acute{\alpha}\hat{r}[4\pi z]$$

$$\frac{\partial r}{\partial t} = 2\acute{\alpha}\hat{r}$$

$$r = 2\acute{\alpha}\acute{K}C_{As}t + C$$

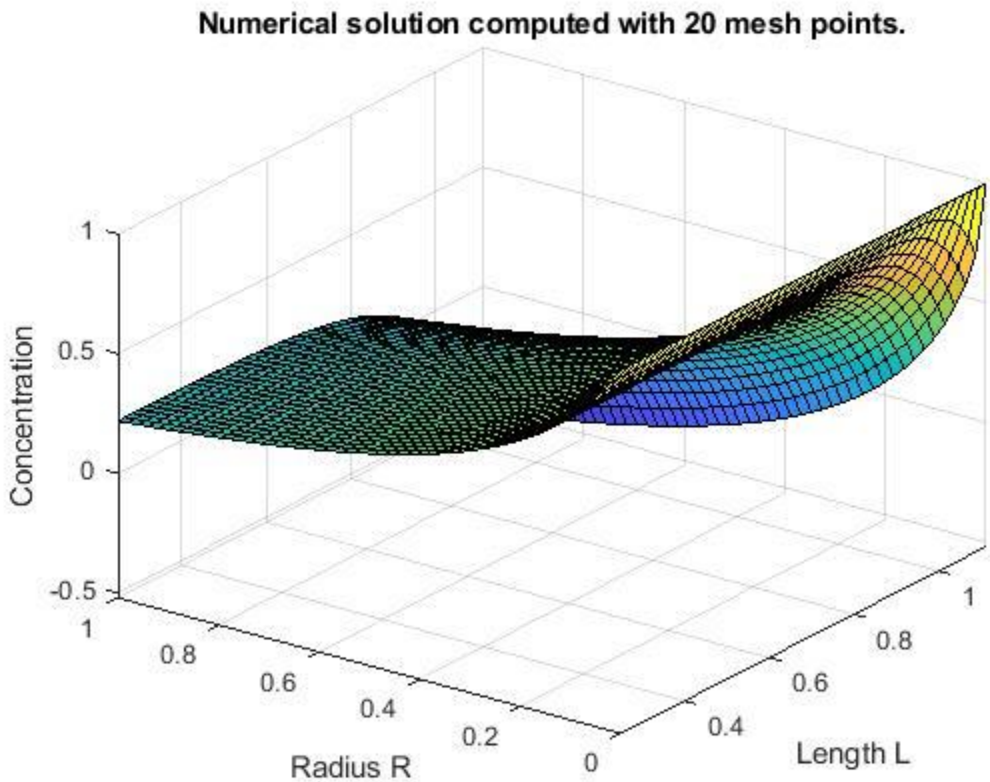
$$t = 0, r = r_0$$

$$r = 2\acute{\alpha}\acute{K}C_{As}t + r_0$$

Part B)

Consider the combustion of the coal. The air **flows at velocity of 1 cm/s..** the **diffusity coefficient of air is 1.6 * 10^{-5} m^2/s and $C_{A,f}$ is 1 mol/m^3. The concentration of the oxygen is 0.2 mol/m^3** .The reaction constant is **0.00009 (cm^2.s)$^{-1}$ at 400 º C . And mass transfer coefficient of air is 1.5*10 ^-3 cm/s.** **The pore diameter at t=0 is 0.5 cm.** and **the volume of 13.3 cm^3 coal /mol O_2** is consumed by air. **plot the profile of reaction rate of the reaction as function of radius and length in 30 minutes of combustion at length of 1 cm . [5 points]**

$$C_{(s)}+O_{2(g)} \longrightarrow CO_{2(g)}$$




```
function r=Homework4
clc
V=1; %cm/s;
r0=0.25; %cm;
DAB=1.6*10^-5*10^4; %Cm^2/s;
CAF=1; %mol/s
CAS=0.2; %mol/s
Kr=0.00009; %1/(Cm^2.s)
alphap=13.3; %cm^3/mol ...1mol O2* (1mol C/1mol O2)* (12 gr C/1molC) *
(10^3 cm^3/900 gr)
```

```
Kc=1.5*10^(-3); %cm/s
t1=30*60;% s
t=linspace(0,t1,50);
%r=2??CAst+r_0
r=2*alphap*Kr*CAS*t+r0;
% (?C_A)/?t+V.[1/r  (??(rC?_A))/?r+(?C_A)/?z]-1/r  ?/?r (rD
(?C_A)/?r)=0
% z=0,C_A=C_Af
% r=0,(?C_A)/?r=0
% r=R,?(C_A-C_(A,) )=?AC_As


m = 1;
z = linspace(0,1,50);
R= r;


sol = pdepe(m,@pdex1pde,@pdex1ic,@pdex1bc,z,r);
% Extract the first solution component as z.
ca = sol(:,:,1);

% A surface plot is often a good way to study a solution.
figure;
surf(z,r,ca)

title('Numerical solution computed with 20 mesh points.')
xlabel('Length L')
ylabel('Radius R')

end
```